VORTEXA 3.0

12 HRS HACKATHON

POWERED BY A.I.R.S

DOMAIN & NO : 7, G-02 Government Systems & GovTech

PROBLEM STATEMENT TITLE : GovFix: Configurable Public-Institution Workflow Engine

TEAM NAME : Brainstormers

TEAM ID : IDD 0506890



PROBLEM STATEMENT

Municipal complaints often move through fragmented, manual channels.

1 REPORTED
Citizen raises issue

2 FORWARDED
Email / spreadsheet

3 WAITING
No clear ownership

4 OVERDUE
No tracked escalation

DELAYED RESPONSE
Cases can remain unattended without time-bound action.

UNCLEAR RESPONSIBILITY
Citizens do not know who owns each grievance.

UNCONTROLLED STATUS
Invalid or unauthorized changes can go unnoticed.

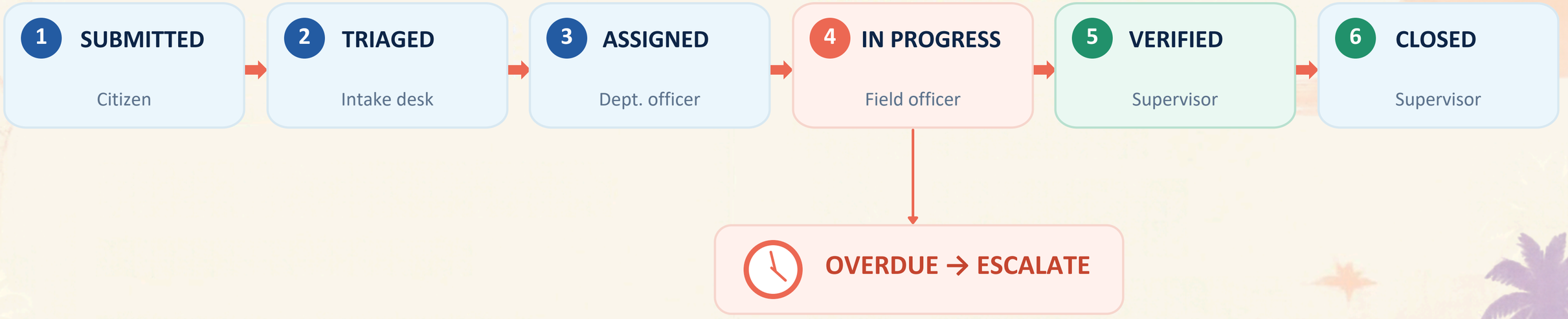
INCOMPLETE AUDIT
Assignments and refusals may leave no reliable trail.

G-02 CHALLENGE · Enforce valid states, authorized roles, deadlines, escalation, and traceable decisions.

IDEA TITLE / PROPOSED SOLUTION

GovFix: a rule-governed grievance journey with AI as an assistant, not the authority.

SIX-STATE WORKFLOW



01 AI-SUPPORTED INTAKE

Suggest category and extract details; human reviews suggestions.

02 ENFORCED ROLE + STATE

Backend blocks skipped states and unauthorized actions.

03 AUDITABLE CLOSURE

Log assignments, refused attempts, deadlines, and escalations.

TECHNICAL APPROACH

Layered architecture: AI recommends; deterministic policy checks every action.

USER INTERFACES

Citizen portal · Staff desk
React frontend

FASTAPI WORKFLOW ENGINE

Role-based access · State machine
Deadline and escalation rules

DATA + IDENTITY

Firebase Auth · Firestore
Complaints + audit events

AI ADVISORY SERVICE

Category, key fields, draft reply
Local deterministic fallback

AUTHORIZED DECISION PATH

Proposed action → Check role → Check state → Record decision

INVALID / UNAUTHORIZED

Block action, explain why, and log refusal.

VALID & AUTHORIZED

Commit transition and preserve who / what / when.

Guardrail: AI suggestions never directly change status, assignment, or SLA deadline.

FEASIBILITY AND VIABILITY

Designed for municipal adoption: reliable rules, accountable operations, and scalable configuration.

THREE PILLARS OF REAL-WORLD VIABILITY

GovFix is built around a reusable workflow—not a collection of disconnected dashboards.

01 TECHNICAL FEASIBILITY

- Established web stack and structured data
- Backend rules validate roles and transitions
- AI is advisory; core flow has a safe fallback

02 OPERATIONAL VIABILITY

- Clear Citizen → Officer → Supervisor roles
- Visible case ownership and status history
- Recorded assignment and overdue escalation

03 ECONOMIC & SCALING VIABILITY

- No mandatory paid AI service or hardware
- Configurable departments, categories and SLAs
- Same engine can serve additional wards

REAL-WORLD RISKS & BUILT-IN SAFEGUARDS

AI CLASSIFICATION ERROR	Suggestions stay editable; no AI-driven state changes.
UNAUTHORIZED UPDATE	API checks role and state; refused actions are logged.
MISSED DEADLINE	SLA checks trigger documented supervisor escalation.
DATA & ACCOUNTABILITY	Role-scoped access and actor-stamped audit events.

CONFIGURE POLICIES AND DEPARTMENTS · KEEP THE SAME TRUSTED WORKFLOW ENGINE

IMPACT AND BENEFITS

Expected operational improvements; outcome measures will be validated in the working demo.

BEFORE · MANUAL FOLLOW-UP

Scattered requests
Unclear owner → Untracked delay

LOW VISIBILITY / INCONSISTENT ESCALATION

WITH GOVFIX · STRUCTURED ACTION

Assigned owner + valid state
Deadline → Escalation → Audit

TRACEABLE, RULE-GOVERNED CASE HANDLING

1 **ASSIGNMENT TIME**
Elapsed time from intake to owner

2 **OVERDUE CASES**
Open requests past their due time

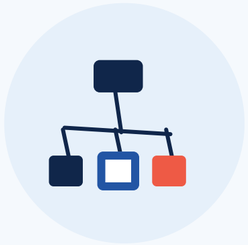
3 **POLICY VIOLATIONS**
Rejected unauthorized / invalid moves

4 **AUDIT COVERAGE**
Actions and refusals with history

Success is demonstrated through observable system behavior—not invented impact percentages.

DIFFERENTIATOR

What makes GovFix distinct.



Rule-Governed Workflow

Only valid state transitions are allowed; skipped or invalid status changes are blocked.



AI + Deterministic Control

AI assists with complaint categorization and response drafting, but cannot change status, assignment, or deadlines.



Deadline-Driven Escalation

Overdue requests are automatically flagged and escalated according to municipal workflow rules.



Role-Based Accountability

Citizens, officers, and supervisors have clearly restricted actions, and refused attempts are recorded.



Complete Audit Trail

Every accepted change, refusal, assignment, and escalation is recorded with who did what and when.



Not just complaint tracking — but accountable, enforceable grievance resolution.

THANK YOU!

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AIRS

Game's Still On

